Ricci-defect variation at a rational-ball end

Exact inner minimization and a cutoff-dependent comparison

Research continuation

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The target remains a smooth closed global minimizer of

$$P(g) = \int_M |E_g|^2 \, dV_g, \qquad E_g = \mathrm{Ric}_g - \tfrac{1}{4} s_g g,$$

which is not Bach-flat. This note computes two precise parts of the remaining rational-ball smoothing problem. It does not establish smooth attainment. The first calculation is an exact minimum for a prescribed quadratic asymptotic jet. The second includes an entire cutoff annulus and shows why an isolated negative finite part is not the insertion coefficient of a fixed critical orbifold.

1 The zero-energy core and its quadratic response

Let

$$G = 1 - r^{-4}, \qquad g_0 = G^{-1} dr^2 + r^2(\sigma_1^2 + \sigma_2^2 + G\sigma_3^2), \qquad r \geq 1.$$

The forms are normalized so that $\sum \sigma_i^2$ is the unit round three-sphere metric. Take the usual free involution of EH covering the antipodal map of the bolt. Its quotient is the ordinary oriented rational ball B_2, with link $S^3/\mathbb{Z}_4$. Write

$$L = \mathrm{Vol}(S^3/\mathbb{Z}_4) = \pi^2/2.$$

The marking and quotient action are fixed throughout. The metric is Ricci-flat and anti-self-dual. More generally, an ALE scalar-flat Kähler rational homology ball is Ricci-flat, and has zero mass [3, 4]. Thus the variable scalar-flat Kähler mass available on other fillings is absent in this class.

Consider the biaxial variation

$$g_t = H^{-1} dr^2 + r^2(\sigma_1^2 + \sigma_2^2 + F\sigma_3^2), \quad F = G + tu + O(t^2), \quad H = G + tv + O(t^2).$$

At a fixed bolt, the linear smoothness conditions are

$$u(1) = v(1) = 0, \qquad u_r'(1) + v_r'(1) = 0.$$

Higher order boundary corrections impose $F'(1)H'(1) = 16$. They do not change the quadratic P coefficient on a fixed compact region, since $E_{g_0} = 0$. Put $x = r^2$ and set

$$\begin{aligned} u &= \frac{\nu - \mu}{6} x + \frac{\nu + \mu}{2x} - \frac{2\nu + \mu}{3x^2}, \\ v &= u - \nu(x - x^{-1}). \end{aligned} \tag{1}$$

This is a smooth quotient-invariant linear response. In the curvature operator convention $R^+ = W^+ + sI/12$, its first variation satisfies

$$DR^+[h] = \mathrm{diag}(\nu, \nu, \mu), \qquad Ds[h] = 4(2\nu + \mu), \qquad h = \dot{g}_0.$$

The homogeneous quadratic jet is Einstein at the cone vertex: $\mathrm{Ric}(p) = (2\nu + \mu) g(p)$. This restriction is stronger than mere $\mathbb{Z}_4$ invariance; arbitrary trace-free Ricci jets are not included.

Proposition 1 (Exact inner minimum)**.** *For the quadratic jet specified by* (1)*,*

$$DE_{g_0}[h] = \mu o, \qquad o = r^{-4}\,\mathrm{diag}(1,-1,-1,1)$$

in the g_0-orthonormal frame. In particular,

$$\int_{B_2} |DE[h]|^2 = \frac{\pi^2}{2}\mu^2. \tag{2}$$

If k is a smooth quotient-invariant correction with $k = O_2(r^{2-\delta})$, $\delta > 0$, and $DE[k] \in L^2$, then

$$\int_{B_2} |DE[h+k]|^2 = \frac{\pi^2}{2}\mu^2 + \int_{B_2} |DE[k]|^2. \tag{3}$$

Thus (2) *is the minimum in this specified class, including non-biaxial corrections.*

Proof. Direct calculation of sectional curvatures gives

$$D\,\mathrm{Ric}[h] = (2\nu+\mu)I + \mu r^{-4}\,\mathrm{diag}(1,-1,-1,1).$$

For example, with x derivatives,

$$\begin{aligned} K_{01} &= K_{02} = -H_x, & K_{12} &= (4-3F-H)/x,\\ K_{13} &= K_{23} = (F-H)/x - HF_x/F,\\ K_{03} &= -H(3F_x/F + 2xF_{xx}/F - xF_x^2/F^2) - H_x(1+xF_x/F). \end{aligned}$$

Substitution of (1) proves the asserted Ricci variation. Since $dV_{g_0} = r^3 dr\, d\omega$,

$$\|o\|_2^2 = 4L\int_1^\infty r^{-5}dr = L.$$

The tensor o is the smooth transverse-traceless scale Jacobi field; $(\nabla^*\nabla - 2\mathring{R})o = 0$, as in the classical Eguchi–Hanson deformation calculation [2]. Pairing $DE[k]$ with o removes the gauge and trace terms. Green's identity gives zero: there is no bolt boundary, and the boundary at radius R is $O(R^{-\delta})$. Expanding the square proves (3). □

There is no logarithmic divergence in (2). Under an inner expansion $g_a = a^2(g_0 + a^2 h + O(a^4))$, it contributes $a^4 L\mu^2$ on expanding rescaled cores, subject to matching estimates. It is not a complete closed-manifold expansion. When $\mu = 0$, this quadratic obstruction vanishes; it supplies no strict exclusion of that degeneration.

2 An angular mode: inner finite part versus the full cutoff

Use

$$\sigma_1^2 + \sigma_2^2 = \tfrac{1}{4}(d\theta^2 + \sin^2\theta\, d\phi^2), \qquad \sigma_3 = \tfrac{1}{2}(d\psi + \cos\theta\, d\phi).$$

A harmonic Gibbons–Hawking moment coordinate is

$$f = \sqrt{r^4-1}\,Y, \qquad Y = \sin\theta\cos\psi, \qquad f_0 = r^2 Y.$$

Choose the free involution preserving this moment coordinate; f then descends to B_2. The angular average is $\mathrm{av}\,Y^2 = 1/3$. The vertical-angle term is essential: the radial equation is

$$r^{-3}(r^3 GA')' - r^{-2}\{8 + 4(G^{-1}-1)\}A = 0, \qquad A = \sqrt{r^4-1}.$$

One obtains

$$\operatorname{av}|df|^2 = 4r^2 - \frac{4}{3r^2}, \qquad \int_{r\le R}|\nabla^2 f|^2 = L\left(4R^4 - \frac{8}{3} - \frac{4}{3R^4}\right)$$

by the Ricci-flat Bochner identity. For $h_f = 2fg_0$, $DE[h_f] = -2\nabla^2 f$, and hence

$$\lim_{R\to\infty}\left\{\int_{r\le R}|DE[h_f]|^2 - \int_{r\le R}|DE_{\text{flat}}[2f_0 g_{\text{flat}}]|^2\right\} = -\frac{32L}{3} = -\frac{16\pi^2}{3}. \tag{4}$$

At the same time $Ds[h_f] = 0$ and $DW^+[h_f] = 0$. The finite part (4) therefore cannot be identified with the chiral bulk integral while discarding its boundary term.

This can be checked with an entire smooth cutoff, not merely the inner ball. Let $\chi(\rho) = 1$ near $\rho = 0$ and $\chi = 0$ for large ρ, and define $f_R = \chi(r/R)f$ and $f_{0,R} = \chi(r/R)f_0$. For compactly supported conformal changes of either Ricci-flat model, Chern–Weil and the scalar transformation law give exactly

$$P(e^{2tu}g_0) = 3\int(t\Delta u + t^2|du|^2)^2\, dV_{g_0}.$$

Here $P(g_0) = 0$, and the conformal change is identical outside a compact set, so the boundary correction does not change.

Proposition 2 (Complete cutoff comparison)**.** *Let $[t^2]$ denote the coefficient of t^2, rather than the second derivative. Then*

$$\lim_{R\to\infty}[t^2]\{P(e^{2tf_R}g_0) - P(e^{2tf_{0,R}}g_{\text{flat}})\} = -3L\int_0^\infty(\rho^3(\chi'')^2 + 3\rho(\chi')^2)\, d\rho. \tag{5}$$

The right side depends on the cutoff profile.

Proof. Write $B_0 = \rho^2\chi'' + 7\rho\chi'$. The exact radial product rule gives

$$\Delta(\chi(r/R)f) = \left(AG\chi_{rr} + A(7/r + r^{-5})\chi_r\right)Y.$$

On the fixed support of the derivatives of χ, this equals

$$\left[B_0 + R^{-4}\left(-\tfrac{3}{2}\rho^{-2}\chi'' - \tfrac{5}{2}\rho^{-3}\chi'\right) + O(R^{-8})\right]Y.$$

Both models have radial volume $r^3 dr\, d\omega$. The limit of the difference of their squared-Laplacian integrals is

$$\frac{L}{3}\int\{-3\rho^3(\chi'')^2 - 26\rho^2\chi''\chi' - 35\rho(\chi')^2\}\, d\rho = -L\int\{\rho^3(\chi'')^2 + 3\rho(\chi')^2\}\, d\rho.$$

Multiplication by 3 proves (5). □

With $z = \log\rho$ and $\psi(z) = \chi(e^z)$, the profile integral is

$$\int_{\mathbb{R}}((\psi'')^2 + 4(\psi')^2)\, dz.$$

If $\psi(z) = \psi_1(z/T)$ is a transition of logarithmic width T, this becomes $T^{-3}\|\psi_1''\|_2^2 + 4T^{-1}\|\psi_1'\|_2^2$. It tends to zero as $T\to\infty$. In particular the isolated number in (4) is not the answer to the fixed-bulk insertion problem. The backgrounds in (5) are being varied with χ and R; they have not been required to be closed P-critical metrics.

3 The remaining closed-manifold comparison

The inner minimization and the cutoff calculation answer distinct parts of the same question. A closed P-critical orbifold has a fixed exterior geometry. Its Euler–Lagrange equation removes the interior first-variation pairing, but a singular ALE tail leaves a boundary concomitant. The subleading inner tail, angular components, and higher exterior jets must be matched before the finite a^4 coefficient has an invariant meaning. Neither (2) nor (4) has determined that coefficient.

There is a concrete reason not to set that boundary term to zero using stationarity. The leading flat-coordinate Eguchi–Hanson tail is

$$h_{ij}^{\infty} = \frac{x_i x_j - (Jx)_i (Jx)_j}{r^6} = \tfrac{1}{8}(\partial_{ij} - J_i{}^k J_j{}^l \partial_{kl}) r^{-2}.$$

Its distributional linearized Ricci defect pairs with a trace-free, divergence-free test tensor through the contraction $J_i{}^k J_j{}^l \partial_{kl} E_{ij}(0)$, whereas the flat principal critical equation controls ΔE. These are different contractions. For example, take $Je_1 = e_3$, $Je_2 = e_4$ and

$$E = (x_1^2 - x_2^2)\,\mathrm{diag}(0, 0, 1, -1).$$

Then $\operatorname{tr} E = \operatorname{div} E = \Delta E = 0$, but $J_i{}^k J_j{}^l \partial_{kl} E_{ij} = 4$. It is generated by the TT quartic metric jet $h_4 = -(x_1^4 - x_2^4)\,\mathrm{diag}(0, 0, 1, -1)/6$ through $DE[h_4] = -\Delta h_4/2 = E$. This is a statement about the principal jet equations, not a construction of a global critical metric. It identifies information that those equations alone do not eliminate.

Two existing tools do not supply this missing comparison. With tensor Weyl norm, set $\mathcal{B}_t = \int |W|^2 + t \int s^2$. Then

$$2P = \mathcal{B}_{1/6} - 32\pi^2 \chi(M).$$

At a closed $\mathcal{B}_t$-critical metric with $t \neq 0$,

$$\nabla P = (t - 1/6) s E.$$

Thus the non-Einstein gluing constructions at the different parameters in [5] do not furnish P-critical metrics. Also, for an order-four Ricci-flat ALE end, inversion gives

$$\bar{g} = \delta + O_2(\rho^4), \qquad W_{\bar{g}}(p) = 0.$$

The compactification-point Weyl interaction in [6] therefore vanishes for this end. Higher asymptotic curvature interactions remain, and any comparison for P must include the scalar-square term as well.

The unresolved statement is a strict closed-manifold energy comparison against the singular minimizing configurations that can actually occur, together with control of the other compactness alternatives. No smooth global non-Bach-flat minimizer has been produced in this continuation.

References

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